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2D

Option 1:
I could have used nested IF statements instead of the case statement, but doing so would have increased the complexity of the synthesis. A case statement is essentially a Mux - an if statement is not as simple.

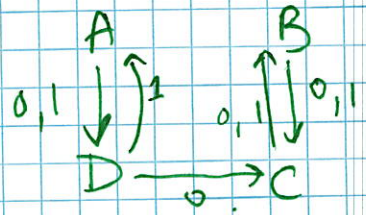
Option 2:
I could have written the case statement to take only 'taken' or only 'cs' as an input, and created nested case statements. this is redundant - ~~wha~~ despite the code's logic being more readable, it is less efficient (if it isn't synthesised out!).

Option 3:
I could have ~~taken~~ the inputs & current state values ~~to create~~ and computed the boolean logic gate circuits needed to arrive at the output state & prediction. This would have taken me far more time to write, which is a tradeoff in itself.
(did 2c ~~before~~ 2B)

(2e)

Group 1: B, C.

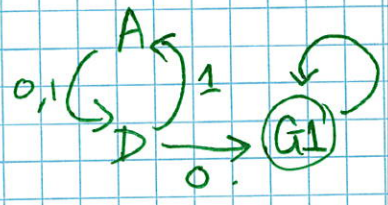
Group 2: A, D.



For group 1,

B 0 → C 1
B 1 → C 0
C 0 → B 1
C 1 → B 1

} G1 to G1. same.



For group 2,

A 0 → D 1
A 1 → D 1
D 1 → A 1
D 0 → G1 ← different.

Change G2 to be just A and G3 to be just D.

for group 1, G1 → G1.

for group 2, G2 → G3 + one state group.

for group 3, G3 → G2
→ G1

but it's fine as a 1 state group.

State	input 0	input 1
A	D1	D1
X	XX	XX
D	A1	XX.

input (Don't care = X).

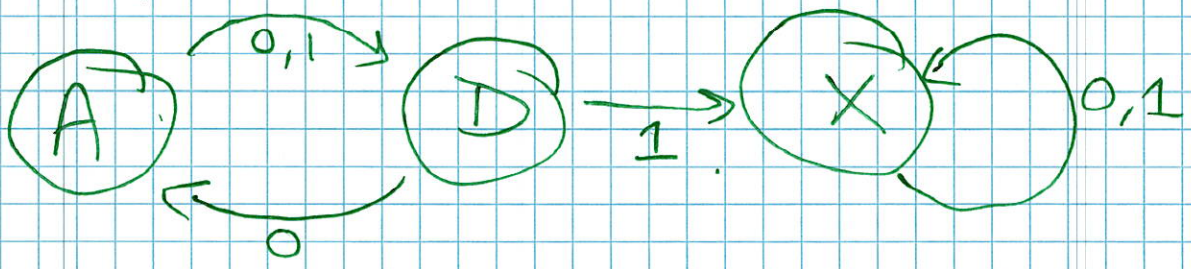
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(2f)

State	input 0	1
A	D,1	D,1
X	X,X	X,X
D	A1	X,X.



This is a Mealy machine - all outputs are functions of the current state combined with an input.